LionX compute: Amazon ECS

ADR-0001. The compute platform decision behind the AWS migration.

ECS Fargate

ECS on EC2

EKS

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The decision, and what drives it

One choice to make: where LionX compute runs on AWS.

THE WORKLOAD

8 services
long-running, plus cron

8 vCPU, 32 GB
per environment

5 environments
integration to production

Networking, CI/CD, observability and IAM all hang off this choice.

WHAT DRIVES THE DECISION

1. **End-2026 cutover.** The platform has to fit the migration runway, not the other way around.
2. **A rewrite either way.** The deployment layer is rewritten regardless. The only question is the target.
3. **Audit surface stays minimal.** Every extra component is more to audit, scan, log and upgrade.
4. **A single workload.** No co-tenants to amortize a cluster's overhead against.

The framing fact

What LionX actually uses today.

LionX uses no Kubernetes feature.

~~operators~~ ~~CRDs~~ ~~service mesh~~ ~~StatefulSets~~ ~~HorizontalPodAutoscaler~~

The current on-prem cluster is a delivery vehicle, not an architectural commitment.

Nothing in LionX pays for the Kubernetes API surface. Choosing Kubernetes on AWS would keep a cost whose benefit we never collect.

Three options

Same workload on each, judged on what costs engineer time.

	ECS Fargate	ECS on EC2	EKS standard to Auto Mode
KUBERNETES TO RUN	none	none	cluster, upgrades, GitOps
NODE OPERATIONS	AWS's job	ours: AMIs, patching, CVEs	ours, or AWS's with Auto Mode
AUDIT SURFACE	CloudTrail only	plus OS-level evidence	plus Kubernetes audit logs
FIT FOR LIONX	runs everything, nothing unused	runs everything, ops attached	pays for features nothing uses
FULLY LOADED, PER YEAR	**~$8K to 11K**	**~$18K to 33K**	**~$27K to 95K**

The cheapest bill buys the most ops. Numbers on the next slide, full pros and cons in the ADR.

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The numbers

Yearly, for the LionX footprint. Zurich list prices with a 1-year Savings Plan, ops deliberately underpriced at $1,000 a day.

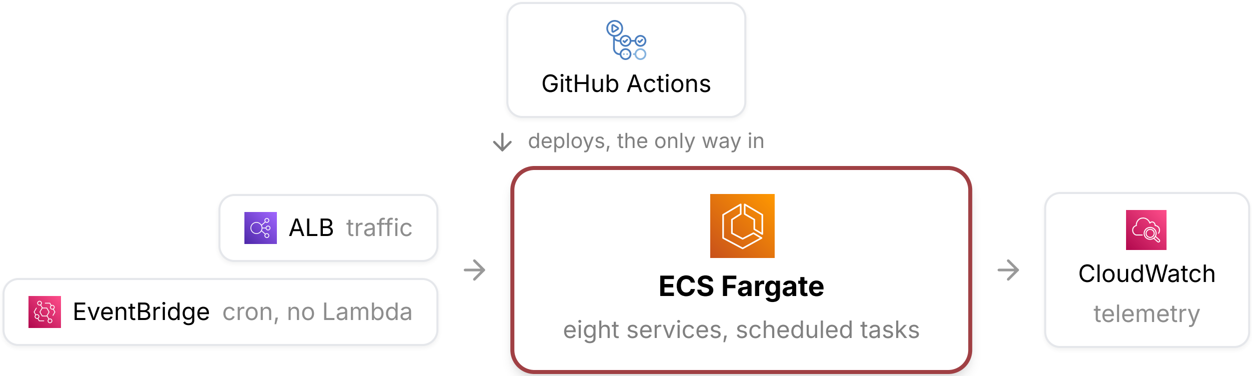
	ECS Fargate	ECS on EC2	EKS standard to Auto Mode
INFRA, PER YEAR	~$4.6K	~$3.1K	~$4.3K to 5.2K
OPS, HOURS PER MONTH	~2 to 4	~10 to 20	~15 to 60
OPS, PER YEAR	~$3K to 6K	~$15K to 30K	~$23K to 90K
FULLY LOADED, PER YEAR	**~$8K to 11K**	**~$18K to 33K**	**~$27K to 95K**

Fully loaded, Fargate is the cheapest option, not the priciest. Ops hours decide it, the EKS range sits roughly 3 to 12 times above Fargate.

eu-central-2 list prices, AWS Pricing API, July 2026. Ops hours are assumptions from industry write-ups, detailed per EKS variant in the ADR.

Decision: ECS Fargate

The smallest operational surface that runs all of LionX.



Zero Kubernetes to run
no cluster, no CNI, no GitOps reconciler, no upgrade calendar

One audit surface
CloudTrail end to end, host patching is AWS's responsibility

Nothing blocks landing
no cluster standup on the critical path to end 2026

Accepted trade-offs. Each schedule is an EventBridge rule plus an IAM role, more Terraform than a CronJob. The ecosystem is narrower than Kubernetes, and nothing we run needs the difference.

Guardrails and next steps

How we keep the decision honest.

CONFIRMATION

- Every workload on Fargate, audited in the migration playbook. Nothing escapes to raw EC2 or EKS.
- CI is the only deployer. Manual service updates are flagged at PR review.
- CloudTrail retention matches Pictet audit policy.

FOLLOW-UPS

- PoC in the new AWS account: pipeline, IAM boundaries and the cron path, end to end.
- Revisit if phase 2 brings a hard need for mesh, operators or stateful Kubernetes.
- Revisit if a shared EKS platform lands at Pictet with co-tenants to split the cost.

Status: proposed. Comments on the PR, then we ratify.